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A REVIEW PAPER ON DEV-G: REVOLUTIONIZING EDTECH WITH MERN AND WEB3 INTEGRATION

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ABSTRACT

The growth of online education has introduced platforms that prioritize accessibility and convenience. However, these platforms often overlook critical aspects such as content ownership, student accountability, and trust in certification processes. DEV-G, a novel ed-tech solution, leverages the MERN stack and Web3 technologies to address these gaps. The platform integrates blockchain for decentralized certification and ownership, smart contracts for transparent monetization, and AI for personalized learning and engagement tracking. This paper delves into the architecture, implementation, and unique capabilities of DEV-G, offering a comparative analysis against existing platforms while exploring its potential to redefine the ed-tech landscape.

Keywords: EdTech, Blockchain, Web3, Decentralized Certification, Smart Contracts, MERN Stack, Artificial Intelligence.

PLANNING OF THE PROJECT

Stage	Period
Research	3 weeks
Feasibility Study	1-2 weeks
System Design	2-3 weeks
Proof of concept Development	1 month
Blockchain platform selection	2-3 weeks
Integration	1 month
User Interface Design	1 month
Business Process Redesign	2 months
Testing	1 month
Pilot Program Design	2 weeks
Execution	1 month

I. INTRODUCTION

Education technology (ed-tech) has evolved to meet the demands of learners worldwide by making education accessible and interactive. However, existing platforms often fail to resolve significant challenges, such as **content ownership rights, accountability in learning,**

- DEV-G aims to redefine the ed-tech landscape by addressing these limitations through the integration of **Web3 technologies** and **AI-driven systems**. The platform leverages blockchain for tamper-proof and decentralized certifications, ensuring authenticity and ownership.
- AI mechanisms enable personalized learning paths and engagement tracking to hold students accountable for their progress. Additionally, smart contracts are used to automate monetization, ensuring fairness and transparency for educators and learners alike.

This paper explores the system architecture, development lifecycle, and unique features of DEV-G, offering a comprehensive analysis of its benefits compared to traditional platforms.

II. COST BENEFIT ANALYSIS

To evaluate the feasibility of DEV-G, a cost-benefit analysis was conducted, comparing traditional ed-tech platforms to blockchain-integrated solutions

A. Cost of Implementation

Blockchain integration introduces additional costs for development and maintenance, including smart contract deployment and blockchain hosting fees.

B. Benefits of Integration

The blockchain-enabled features significantly outweigh the costs by providing:

1. **Transparency:** Automated smart contracts reduce operational overhead.
2. **Authenticity:** Blockchain certificates eliminate fraudulent claims.
3. **Ownership:** Educators benefit from transparent revenue sharing.

III. LITERATURE SURVEY

1. Current State of EdTech

A variety of platforms dominate the ed-tech space:

Platform	Weaknesses
Udemy	Centralized, lacks engagement enforcement
Coursera	Expensive, centralized certifications
OpenCerts	Limited to credentials, no learning ecosystem

The reliance on centralized models has resulted in reduced trust, particularly in the areas of certification and ownership.

2. Blockchain in Education

Blockchain applications in education are still in their infancy. OpenCerts, for instance, offers decentralized certification but does not integrate other features like engagement tracking or revenue sharing.

3. AI in Education

AI is primarily used in ed-tech platforms for recommendation systems (e.g., suggesting courses). However, its potential for improving engagement and enforcing structured learning remains largely untapped.

4. Research Gap

While blockchain and AI have been explored separately, their integration with core ed-tech functionalities has not been realized. DEV-G fills this void by combining these technologies to offer a robust, learner-centric platform.

IV. Methodology

1. Research Design

The research involved:

- Identifying limitations in existing ed-tech platforms through surveys and literature analysis.
- Developing a prototype of DEV-G with MERN and Web3 technologies.
- Testing the platform with a cohort of students and educators to evaluate its effectiveness.

2. Technical Architecture

Frontend

- **React.js:** Used to create an intuitive user interface with dynamic components.
- **Tailwind CSS:** Provides rapid prototyping and ensures responsiveness across devices.

Backend

- **Node.js and Express.js:** Handle API requests, user authentication, and business logic.

Database

- **MongoDB:** Stores user profiles, course content, and engagement data in a scalable NoSQL structure.

Blockchain Integration

- **Ethereum and Solana:** Implement decentralized storage and certification.
- **Smart Contracts:** Facilitate revenue sharing and certification issuance.

AI and Analytics

- Machine learning models track student engagement metrics, including:

- Video watch duration.
- Quiz performance.
- Interaction frequency.

V. Implementation

1. Workflow

For Educators:

1. Course Creation: Educators upload content and configure settings for assessments.
2. Revenue Sharing: Smart contracts ensure transparent and timely payouts.

For Students:

1. Enrollment: Students register and pay through traditional or crypto payment methods.
2. Learning Path: AI enforces sequential learning by restricting access to skipped modules.
3. Certification: Blockchain-stored credentials are issued upon course completion.

2. Security Features

- **SHA-256 Encryption:** Secures user passwords and sensitive data.
- **HMAC Algorithms:** Enhance transaction security during payment processing.

VI. RESULTS

A group of 50 students and 10 educators used DEV-G for three months. Findings included:

- **Engagement Improvement:** 35% increase in course completion rates.
- **Certification Trust:** Employers expressed confidence in blockchain-based credentials.
- **Revenue Transparency:** Educators reported a seamless payout process.

Comparison with Existing Platforms

Feature	DEV-G	Udemy	Coursera
Decentralized Ownership	✓	✗	✗
AI-Driven Engagement Tracking	✓	✗	✓
Blockchain-Based Certification	✓	✗	✗
Revenue Sharing Transparency	✓	✗	✗

The thesis concludes that understanding the costs and benefits of implementing blockchain-based SCF solutions is crucial for businesses. The cost-benefit model presented in the thesis can be used to quantify the net value of implementing blockchain-based SCF solutions. The results suggest that blockchain technology can increase the total net benefit of involved parties in SCF as a result of improved efficiency of invoice processing.

VII. BLUEPRINT

This comprehensive diagram ensures efficient interaction between students, teachers, and the underlying database structure. The modular design allows for easy scalability and adaptability, ensuring future integration of advanced features such as blockchain-based certificates and AI-driven learning enhancements.

This blueprint serves as a foundational guide to understanding the platform's architecture and its data-driven operations, reinforcing the core objective of DEV-G to deliver a transparent and engaging learning experience.



REFERENCE

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